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Classification of sound environment based on subjective response with speech privacy in open plan offices

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ABSTRACT

To implement pleasant open plan offices (OPOs) in the preference aspect based on the subjective responses of the workers, we proposed novel acoustic environment design classification criteria for speech privacy-related parameters. To this end, we created acoustic stimuli through computer simulations and surveyed the related preference responses of test participants. Then, we selected classification criteria based on the survey response results. First, field measurements were performed in two typical OPOs, and a computer simulation model was built. Twelve different sound stimuli were generated by changing the material absorption coefficient conditions for the ceiling and floor inside an office. The investigation of the noise sensitivity and satisfaction of each sound source according to the individuals indicated that the A-weighted sound pressure level at 4 m ($L_{p,A,S,4m}$) and spatial decay rate of speech ($D_{2,S}$) were the major acoustical parameters that determined OPO satisfaction, with $L_{p,A,S,4m}$ considered first in the OPO design stage. The $L_{p,A,S,4m}$ class interval was set to 2 dB, and a new four-level classification criteria was proposed, in which the participant response rates of satisfaction were divided according to the change of $L_{p,A,S,4m}$ into 25% intervals: A ($\leq$ 46 dB), B (46 dB $\leq x \leq$ 48 dB), C (48 dB $\leq x \leq$ 50 dB), and D ($\geq$50 dB). A large difference of two or more satisfaction classes was observed between the high and low noise sensitivity groups for the same $L_{p,A,S,4m}$, confirming that the noise sensitivity of workers influences the preferences for OPOs. The proposed classification criteria are expected to be applicable to the design of comfortable OPOs in the future.

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1. Introduction

1.1. Sound environment in open plan offices

Open plan offices (OPOs) can enhance teamwork and creativity by improving communication and collaboration efficiency in terms of work. Further, OPOs can reduce construction and maintenance costs and ensure free space flexibility. Hence, they have been frequently used [1,2]. Nevertheless, the advantages of OPOs remain controversial. Many studies have reported that the acoustic problems associated with OPOs, such as noise, lack of privacy, and auditory distraction, have negative effects on worker satisfaction or productivity [3–6]. In addition to acoustics, several other factors, such as air quality, thermal comfort, and lighting, determine the indoor environmental quality of OPOs. Among these factors, studies have shown that the acoustic environment has the greatest effects on work productivity and satisfaction [7,8]. In particular, the lack of sound privacy has been demonstrated to be a major acoustic factor negatively affecting the subjective overall workspace satisfaction of workers in OPOs [7,9]. Intelligible conversation among workers or irrelevant speech were determined to be the primary causes of decreased worker satisfaction and performance and increased mental workload in OPOs [10–20]. Therefore, the violation of privacy due to worker speech must be considered when designing the sound environment of an OPO to improve worker satisfaction.

1.2. Physical parameters for speech privacy in open plan office

The concept of speech privacy is inverse to that of speech intelligibility and is defined as the inability of an unintentional listener to understand the conversations of other people. Various physical indicators to quantify speech privacy in OPOs have been suggested [21–24], such as the articulation index (AI) based on the signal-to-noise ratio by frequency band, which is normalized to a value between 0 and 1 [21,25]. In general, an OPO can be considered to provide confidential privacy when the AI is 0.05 or lower [26] and acceptable privacy when the AI is 0.15 or lower [22]. Another indicator proposed is the speech intelligibility index (SII), which is

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similar to the AI but considers the voice spectrum change according to vocal effort [27]. Confidential privacy can be ensured if the SII is 0.1 or lower, whereas the SII must be 0.20 or lower to ensure acceptable privacy [28]. Further, the speech transmission index (STI), which is a physical quantity representing the quality of voice transmission related to articulation, has been proposed. The STI is a measure of the degree of distortion that determines the sign of intelligibility by assuming that the main cause of low intelligibility of voice in an indoor space is the distortion of the original sound [23]. The STI has a range of 0–1, and generally, the higher the STI, the higher the intelligibility and the lower the speech privacy [29]. Hongisto [30] verified that the performance decreases in the STI range of 0.2–0.6 and that the appropriate level of the STI is approximately 0.2 to minimize work obstruction due to surrounding speech. A follow-up study [31] concluded that an STI of 0.4 or lower can ensure reasonable speech privacy. The rapid speech transmission index (RASTI), which simply calculates the STI, has also been suggested. The RASTI is a subjective indicator of speech intelligibility according to voice transmission in a room, considering the background noise in the room [24]. However, the RASTI does not consider distortions in the time domain. Many other indicators for evaluating and quantifying speech privacy have been proposed, including articulation loss of consonants, evaluated by the energy ratio of direct and indirect sounds (%ALCons) [32]; definition, evaluated by the ratio of energy until 50 ms after reaching the direct sound to the total energy (D_{50} [33]); and reverberation time, calculated by doubling the decay time in the range from 5 to 35 dB of the energy decay curve after the sound source has stopped playing (RT_{30}, [34]).

The ISO 3382–3 guidelines established in 2012 by the ISO/TC 43/SC 2 Building Acoustics Technical Committee suggest four single-number quantities for ensuring good speech privacy in OPOs [35]. First, $L_{p,A,S,4m}$ denotes the A-weighted voice sound pressure level (SPL) at a distance of 4 m from the sound source. Further, the spatial decay rate of speech $D_{2,S}$ is the decay rate of the A-weighted voice SPL as the distance doubles and is calculated by linear regression with a logarithmic distance axis within 2–16 m. The distraction distance r_D is determined by the distance at which the speech transmission index from the speaker decreases to 0.5 or lower. $L_{p,A,B}$ is the A-weighted background noise level. Many studies have been conducted to evaluate the speech privacy of OPOs using various related single-number quantities [6,36–39]. Virjonen et al. [40] proposed four classes denoted as A–D considering case studies conducted on 16 office spaces based on the $L_{p,A,S,4m}$, $D_{2,S}$, and r_D indicators. They concluded that $L_{p,A,S,4m}$ should be decreased to 47 dB or lower, $D_{2,S}$ to 7 dB or higher, and r_D to 5 m or lower. These values were established by ISO 3382–3 Annex A (2012). Lee and Jeon [38] demonstrated through auditory evaluation that $D_{2,S}$ and $L_{p,A,S,4m}$ are useful for evaluating the listening difficulty for speech and confirmed that r_D is a useful index for evaluating speech intelligibility and listening difficulty.

Haapakangasa and Hongisto [41] also verified the usefulness of $L_{p,A,S,4m}$, r_D, and $L_{p,A,B}$ (average A-weighted SPL of background noise) following this order to evaluate annoyance responses to noise and speech. Therefore, the single-number quantities suggested in ISO 3382–3 are appropriate for evaluating the speech privacy of OPOs. However, although the relationships between these single-number quantities and subjective response (speech intelligibility and privacy) have been revealed, research on appropriate design criteria remains insufficient. Virjonen et al. [40] attempted to prepare sound environment classification criteria for certain OPOs, but they did not reflect actual understanding of workers because the classification was based on simple physical indicators rather than subjective responses.

1.3. Noise sensitivity

Noise sensitivity is a crucial non-acoustical factor that influences annoyance due to indoor and outdoor noise [42–46]. Miedema and Vos [43] examined differences in conversation disturbances according to noise sensitivity and found that the disturbances were higher in the high sensitivity group than in the low sensitivity group. Park et al. [47] reported that the high sensitivity group exhibited a higher noise annoyance reaction to indoor building noise than the low sensitivity group and more distinct changes in physiological responses (heart rate, electrodermal activity, and respiratory rate). Furthermore, the noise sensitivity and effect of noise on health were confirmed to be major factors that influence subjective health complaints (e.g., sleep problems, nervousness, and headaches) [48,49]. Among the studies on OPOs, Lee et al. [50] investigated the relationship between speech privacy and noise sensitivity and discovered that in the high sensitivity group, speech privacy was a critical factor in predicting satisfaction. Abbasi et al. [51] found that among noise exposure, sensitivity, and noise annoyance, noise sensitivity had the greatest effects on job satisfaction and stress. Hence, it can be concluded that the satisfaction with the sound environment varies significantly depending on the noise sensitivity. Nevertheless, when the relationships between the acoustical parameters related to speech privacy and satisfaction were examined, only a few studies have considered noised sensitivity comprehensively. Therefore, individual differences in noise sensitivity must be considered when preparing design standards for comfortable OPO sound environments in terms of the subjective perceptions of workers.

1.4. Evaluation method for open plan office research

Evaluations of the speech privacy of OPOs can be broadly divided into field surveys and laboratory evaluations. The field survey method for workers in an actual office environment can yield actual response results and offers the advantage of ease of generalization of the results because several people can be surveyed. Thus, most previous studies used this method [7,40,50]. In contrast, the laboratory evaluation method reproduces an office space that is physically similar to the field or replays sounds only through headphones or speakers. Although this method is less realistic, it offers the advantage of producing more controlled results by providing the same stimuli to all subjects. Hence, this method has also been adopted [38,52,53]. In particular, when the number of offices to be evaluated is limited, the method of producing various types of stimuli by changing the physical characteristics of the sound environment based on computer simulation has been used [54–56]. Sarwono et al. [56] investigated changes in speech privacy using a workstation model through computer simulation. Rindel [54] proposed a method of improving speech privacy by changing the screen, absorption design, and background noise inside the OPO. In addition, test sounds produced by computer simulation have been used to evaluate the speech privacy of OPOs, verifying the usefulness of this approach [38,57].

1.5. Aim of this study

Against this background, this study aimed to investigate satisfaction with OPOs based on computer simulation and its relationships with various physical parameters related to speech privacy to prepare OPO sound environment design standards based on subjective responses. Thus, the new research questions addressed in this study are as follows:

What are the major factors influencing satisfaction with the work environment among the physical characteristics for evaluating the OPO sound environment?
What are the design classification criteria and class intervals necessary to ensure appropriate speech privacy in OPOs?
To what degree is the class difference due to worker sensitivity?

2. Methods

2.1. Open plan office

This study generated sound stimuli to be used in a subjective evaluation based on computer simulations. First, the in-situ measurement data required for computer simulation modeling were collected. By arbitrarily modelling the computer simulation, the subjectivity of the researcher may interfere with layouts and furniture layouts. Therefore, we adopted a method of computer modeling actual offices to control these external factors. In this study, two office spaces with different shapes were selected to examine the effects of changes in speech SPL on satisfaction under various conditions. Thus, two office spaces with a typical OPO layout at the center of Seoul were selected. The ground plan and physical features of each office space are summarized in Fig. 1 and Table 1.

Office I is an office space used as a research institute with a relatively large area of 38.0 × 35.0 m, with the interior finished using absorption gypsum boards and carpets. In contrast, Office II is an office used by a business team, having an area of 20.2 × 15.2 m, with the interior finished using absorption gypsum boards and vinyl tiles. To investigate the physical acoustic properties of each OPO, in-situ measurements were performed in accordance with ISO 3382–3 (2012), and the results are summarized in Table 1. At the time of measurement, both offices were furnished, and the ceiling air conditioning system was in operation. Furthermore, measurements were performed during normal working hours in an empty room. As a result, the values of $L_{p,A,S,4m}$, $D_{2,S}$, r_D, and the STI were found to be 51.4 dB, 5.7 dB, 10.8 m, and 0.51, for Office I and 49.4 dB, 4.6 dB, 11.1 m, and 0.51, for Office II, respectively. Furthermore, the reverberation time (RT_{30}) values of the two spaces were 0.2 and 0.6 s on average at 500 Hz and 1 kHz, whereas the D_{50} values of the two spaces were 0.95 and 0.89, respectively.

2.2. Computer simulation

Based on the in-situ measurement results, a computer simulation model was implemented to obtain sound environment characteristics that were similar to those of an actual office space. Three-dimensional modeling was performed using AutoCAD software (version 2016) and SketchUp software (version 2017). The internal design factors of the ceilings, walls, floors, partitions, desks, and drawers were implemented in an identical manner. Further, for the acoustic simulation, ODEON simulation software (version 15) was used, and an absorption coefficient similar to that of an actual material was applied. The same acoustical parameters examined in Section 2.1 were analyzed by specifying the same sound source location and sound-receiving point as those in the in-situ measurements. The results of the comparison with the actual values are shown in Table 1 and Fig. 2 (a) and (b). For Office II, the values of all acoustical parameters differed by 4% or less when compared with the actual values, indicating a good approximation of reality. Further, in Office I, the values of acoustical parameters except RT were similar to the actual measurements. Most of the acoustic parameter values of Offices I and II implemented using computer simulations were similar to in-situ measurements, showing that they were in good agreement. Although some differences were observed for RT in Office I, we determined that this was not a major issue as the selection criteria of sounds created in this study were based on $L_{p,A,S,4m}$.

To implement the sound environments corresponding to various conditions in the two office spaces, 12 models with different acoustic conditions were generated by changing the absorption conditions of the ceiling and floor based on a computer simulation, and the results are shown in Tables 2 and 3. In this study, a wide range of 45.9–55.9 dB for the $L_{p,A,S,4m}$ values was selected by referring to the office case studies of Virjonen et al. [40] and Haapakangas et al. [41], wherein the $L_{p,A,S,4m}$ distributions were 44.3–53.8 and 43.4–57.2 dB, respectively. Further, $D_{2,S}$ and r_D changed simultaneously in the ranges of 2.5–8.1 and 6.1–13.2 m, respectively. The range of STI values (0.37–0.55) was such that it covered a wide range of OPO sound environments, from environments with STIs higher than 0.5, where certain speech in existing studies [29,30,31] was heard well, to environments with STIs lower than 0.4, which can be considered to provide a reasonable level of speech privacy. Moreover, in the computer simulation, the absorption coefficient of the floor material inside the office was set to two values (high = 0.5 and low = 0.05), and the absorption coefficient of the ceiling material was set to three values (high = 0.95, mid = 0.50, low = 0.05). Interestingly, when we examined the changes in RT with and without internal furniture such as office desks or partitions inside the OPO, we found that the RT changes based on the changes in the sound absorption coefficients of the ceiling and the floor were not dramatic regardless of the presence of internal furniture. Here, three pairs of models have almost the same acoustic parameters (cases 1 and 4, cases 2 and 5, and cases 7 and 10). However, as the sound absorption coefficients of the ceiling and the floor are opposite to each other, the effects of the reflected sound appear to be different; thus, these pairs of cases were divided. Consequently, 12 room impulse responses were produced

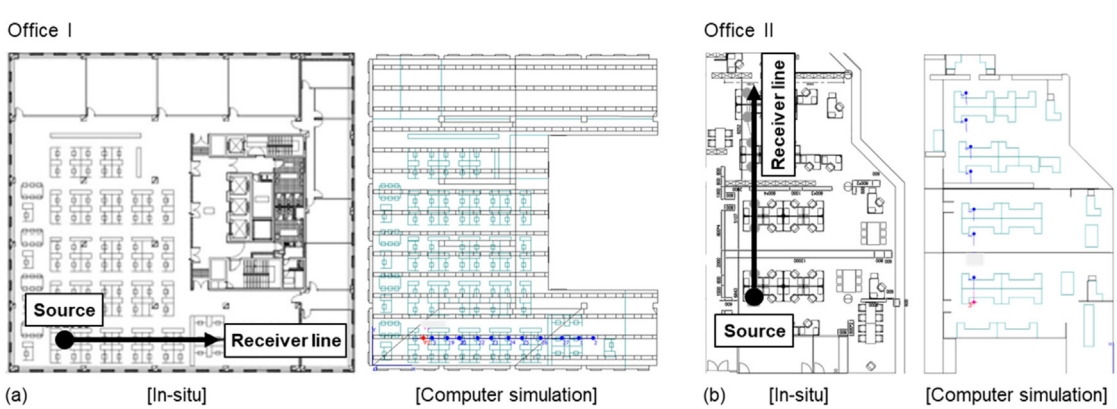

Fig. 1. Ground plan of OPO. (a) Office I and (b) Office II.

Table 1
Physical characteristics of OPO environments.

Site	Office I		Office II	
Office user	Research institute		Business team	
Length × depth	38.0 × 35.0 m		20.2 × 15.2 m	
Ceiling height	2.4 m		2.5 m	
Screen height	1.2 m		1.1 m	
Ceiling/floor	Absorption gypsum board/carpet tile		Absorption gypsum board/vinyl tile	
HVAC	Ceiling air conditioning		Ceiling air conditioning	
Tool	Measurement	Simulation	Measurement	Simulation
$L_{p,A,S,4m}$ [dB]	51.4	49.6	49.4	48.8
$D_{2,S}$ [dB]	5.7	6.2	4.6	4.5
r_D [m]	10.8	9.3	11.1	11.3
STI	0.51	0.47	0.51	0.53
RT [s]	0.2	0.5	0.6	0.6
D_{50}	0.95	0.90	0.89	0.89

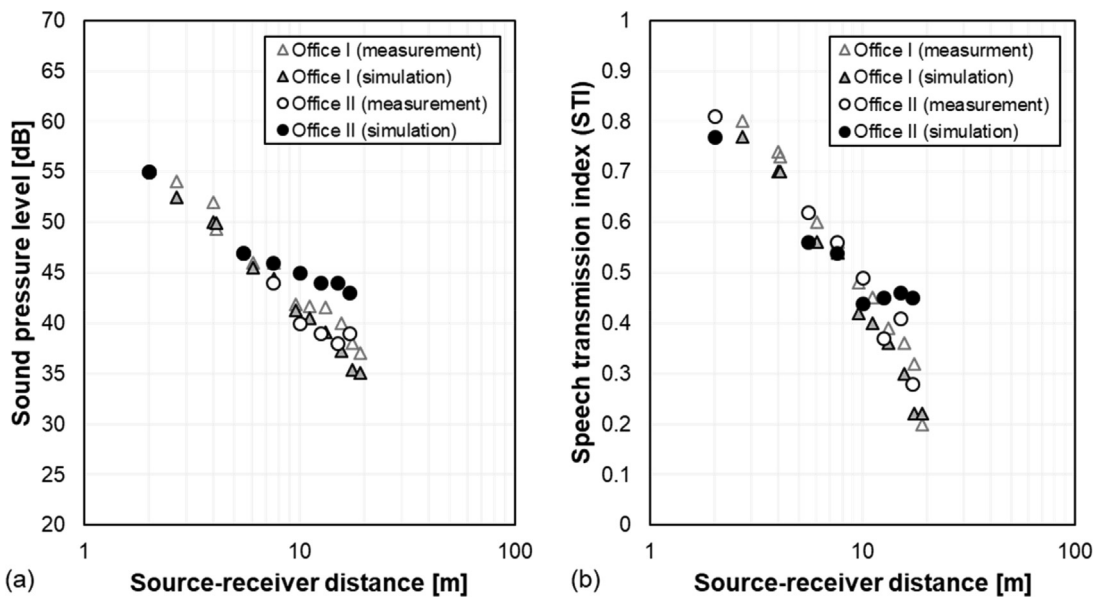

Fig. 2. Comparison of in-situ measurement and computer simulation results for the (a) SPL and (b) speech transmission index according to the source-to-receiver distance.

Table 2
Room acoustic parameters of sound stimuli depending on the absorption conditions.

Case	Site	Absorption coefficient		Room acoustical parameters					
		Floor	Ceiling	$L_{p,A,S,4m}$ [dB]	$D_{2,S}$ [dB]	r_D [m]	STI	RT [s]	D_{50}
1	Office I	High (0.50)	High (0.95)	46.3	8.1	6.8	0.37	0.5	0.81
2			Mid (0.50)	47.0	7.2	7.5	0.41	0.5	0.81
3			Low (0.05)	49.3	5.8	8.5	0.46	0.5	0.74
4		Low (0.05)	High (0.95)	46.6	7.9	6.8	0.38	0.5	0.79
5			Mid (0.50)	47.7	6.8	7.8	0.43	0.5	0.78
6			Low (0.05)	51.3	5.0	8.5	0.46	0.8	0.55
7	Office II	High (0.50)	High (0.95)	45.9	5.2	8.9	0.47	0.5	0.71
8			Mid (0.50)	47.9	4.7	10.1	0.50	0.5	0.72
9			Low (0.05)	51.1	3.9	13.2	0.55	0.7	0.66
10		Low (0.05)	High (0.95)	46.7	5.0	9.1	0.48	0.5	0.65
11			Mid (0.50)	48.8	4.5	11.3	0.53	0.5	0.64
12			Low (0.05)	55.9	2.5	6.1	0.45	1.5	0.28

through computer simulation at the sound receiving point 4 m from the sound source.

2.3. Background noise

The above-mentioned room impulse responses generated using computer simulations do not reflect the background in the field. Hence, the background noise in an actual OPO space was recorded for 1 h for use in the evaluation of sound production, as shown in Fig. 3 (a). The SPL for each octave frequency band of the recorded sound is depicted in Fig. 3 (b). Further, to investigate the background noise level in empty office spaces, the 1-min average L_{Aeq} was calculated for sections excluding the speech elements from the recorded background noise sound, and the average background

Table 3
Absorption coefficients of different materials for modeling using Odeon simulation.

Element	Materials	Frequency								
		63	125	250	500	1000	2000	4000	8000	a
Desk, Drawer, Partition	1-cm-thick plywood paneling (Harris, 1991)	0.28	0.28	0.22	0.17	0.09	0.1	0.11	0.11	0.10
Wall	Brick, unglazed, painted (Harris, 1991)	0.01	0.01	0.01	0.02	0.02	0.02	0.03	0.03	0.05
Floor	Linoleum or vinyl stuck to concrete (Petersen, 1983)	0.02	0.02	0.02	0.03	0.04	0.04	0.05	0.05	0.05
Floor	Carpet, heavy, on hair felt or foam rubber (Harris, 1991)	0.08	0.08	0.24	0.57	0.69	0.71	0.73	0.73	0.50
Ceiling	Plaster, gypsum, or lime, smooth finish on lath (Harris, 1991)	0.14	0.14	0.1	0.06	0.04	0.04	0.03	0.03	0.05
Ceiling	Slotted 13-mm gypsum board (12%), 106 × 3 mm^2 on studs and mineral wool (Ref, Dalenback, CATT)	0.20	0.20	0.22	0.71	0.99	0.55	0.42	0.42	0.50
Ceiling	100-mm-thick wood-wool against the ceiling (Ref. Dalenback, CATT)	0.58	0.58	0.69	0.96	0.97	0.99	0.91	0.91	0.95

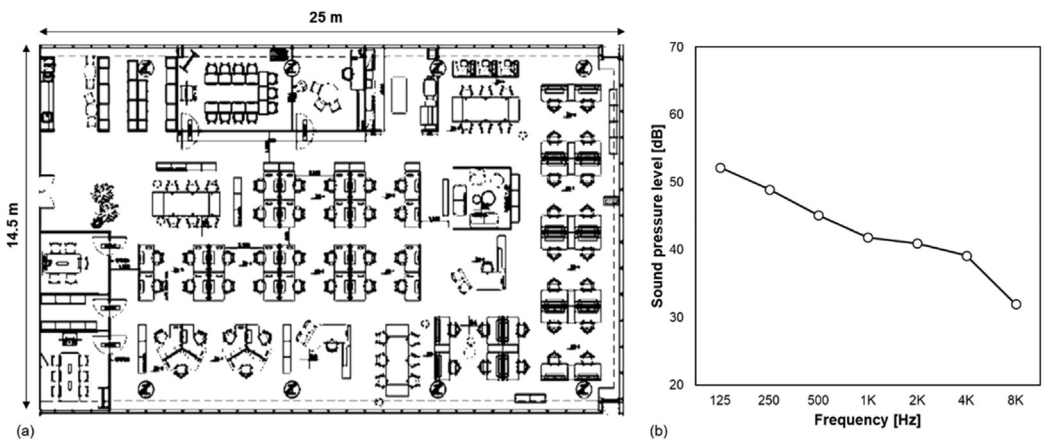

Fig. 3. Background noise characteristics. (a) Measurement setup, and (b) frequency characteristics of background noise.

noise level excluding speech was determined to be 47.0 dB. This value is similar to the average OPO noise level of 46–47 dB obtained in the case study of Seddigh et al. [58]. To generalize these analysis results, background noise data for 1 min were collected from four OPOs in Korea and in English-speaking countries from an open-source dataset (YouTube), and the SPL for each frequency band was analyzed (Table 4) [59]. The results showed insignificant differences in background noise level between Korea and the English-speaking countries, and the background noise level excluding speech was 47.0 dB, similar to the in-situ measurement result. This finding suggests that the background noise level selected in this study represents the background noise level of general unoccupied offices.

2.4. Sound stimuli

The sound stimuli for auditory evaluation were generated based on the binaural room impulse responses produced through computer simulations and the background noise recorded in the field. First, a 1-min speech sound measured in an anechoic chamber was convoluted with the binaural room impulse responses produced above. Subsequently, to ensure that the $L_{p,A,S,4m}$ value of each speech sound was identical to that presented in Table 2, the convoluted sound was played back through headphones (Sennheiser HD 650) and recorded via a head and torso simulator (Type 4100, Brüel & Kjær). The SPL was adjusted using Adobe Audition software (version 1.5). Here, we used the built-in amplitude function of the Adobe Audition software to change the sound pressure level and corrected the $L_{p,A,S,4m}$ values of the created sound stimuli to ensure their consistency with those in Table 2. Finally, 12 sound stimuli were generated by adding the background noises of actual office spaces that were measured in the field. At this time, the signal-to-noise ratios of each sound source stimulus were −0.7 dB (case 1), 0 dB (case 2), 2.3 dB (case 3), −0.4 dB (case 4), 0.7 dB (case 5), 4.3 dB (case 6), −1.1 dB (case 7), 0.9 dB (case 8), 4.1 dB (case 9), −0.3 dB (case 10), 1.8 dB (case 11), and 8.9 dB (case 12).

Table 4
Background noise level of OPOs in the 1/1 octave band [59].

Nationality	Number	Background noise level [dB]							
		125	250	500	1 k	2 k	4 k	8 k	L_{Aeq}
Korea	#1	49.4	52.3	52.5	48.0	46.8	42.7	39.1	54.2
	#2	55.0	53.0	50.0	47.0	46.8	45.9	43.7	54.0
	#3	59.8	53.9	48.3	49.5	45.5	43.0	36.7	53.9
	#4	43.3	50.3	52.8	47.4	46.6	44.7	40.2	54.1
English	#1	46.5	49.3	47.8	50.2	47.5	44.8	39.3	54.2
	#2	48.1	45.9	43.8	46.5	49.4	46.8	40.5	54.0
	#3	50.6	49.5	48.7	47.0	48.3	45.9	45.9	54.1
	#4	50.1	50.4	51.4	45.9	46.8	46.3	44.0	54.0

2.5. Questionnaire

A questionnaire detailing noise sensitivity and preferences was prepared to examine the subjective responses of workers according to the changes in the physical sound environment regarding the speech privacy of OPOs. First, the participants were instructed to evaluate their usual sensitivity to noise on a 7-point Likert scale (-3: not at all, 3: extremely). This study used single-item questions because previous studies derived significant relationships between noise sensitivity and subjective evaluation results by evaluating noise sensitivity utilizing this approach [48–50]. The questions were composed of six items (willingness, annoyance, privacy, immersion, pleasantness, and comfort) related to preferences for each evaluation stimulus. Willingness evaluated the degree to which one wanted to work in an office. Annoyance evaluated the irritation felt regarding the sound environment. Privacy evaluated privacy satisfaction, and immersion evaluated the degree of immersion in the work in a sound environment. Pleasantness and comfort were detailed factors for evaluating the overall satisfaction with the sound environment. The participants were instructed to respond on a 7-point Likert scale (-3: not at all to 3: extremely) regarding the degree of agreement with each evaluation factor.

2.6. Procedure

A total of 37 participants (male = 29, female = 8) aged between 18 and 31 years (mean age: 23.41, standard deviation: 3.56) participated in the survey. The appropriate sample size for multiple linear regression was determined using G-Power software (version 3.1.9.4). It was inferred that 33 people were required for verification with an effect size of 0.2 and a verification power of 0.8. Hence, a participant group of 37 people, larger than the inferred number of people, was selected. In addition, to reduce the variation in the responses of the participants, only students who had experience working in office spaces and were attending the same university were recruited. Graduate school students in their 20 s were considered appropriate as the evaluation participants because they had been working in OPO environments with poor noise conditions for a long time. Here, in the process of recruiting experiment participants, we verbally checked whether they had normal auditory functions to choose only those who had normal hearing. Before the experiment, instructions were provided to the participants, and their written consent to participate was obtained. The laboratory was a low noise environment with a background noise level of approximately 30 dBA.

A total of 12 stimuli, each with a duration of 1 min, were applied to the participants in a random order through open-type headphones (Sennheiser HD 650). After listening to a sound stimulus for 1 min, the participants were instructed to answer the questionnaire with six items. The sound stimuli were repeatedly applied upon participants request. Consequently, 444 response data (12 stimuli $\times$ 37 participants) were collected for each evaluation item.

2.7. Data analysis

In this study, the following analyses were performed using SPSS Statistics (version 25) and R (version 3.5.1). First, principal component analysis (PCA) was performed for the responses provided by the participants, and various preferences for the OPO sound environment were dimensionally reduced to a single satisfaction scale. In addition, multiple linear regression and probit regression analyses were performed to examine the relationships between the physical sound environment parameters and satisfaction scale. Thereafter, the acoustical parameters and criteria for classification were selected based on the contribution of each parameter. Finally, probit analysis was performed between the selected acoustic parameters and satisfaction, and the classification criteria for OPO sound environments were proposed based on the analysis results.

3. Results

3.1. Satisfaction with sound environment in OPOs

To dimensionally reduce the various responses (willingness, annoyance, privacy, and immersion) of participants regarding preferences to a single indicator, PCA was performed for six evaluation factors. Varimax rotation was applied in the PCA, with a value higher than 1 considered as the reference eigenvalue. The Kaiser-Meyer-Olkin measure of sampling adequacy was appropriate at 0.92, and the Bartlett test of sphericity yielded statistical significance (p < 0.01). Consequently, one cognitive scale was derived and referred to as "satisfaction" considering its correlations with willingness (0.884), annoyance (-0.875), privacy (0.802), immersion (0.944), pleasantness (0.912), and comfort (0.915); this scale had an explanatory power of 79.1% of the total variance. Further, the factor score was calculated via regression analysis of the satisfaction derived through PCA. Moreover, Pearson correlation analysis was performed to investigate the correlations between seven preference attributes, including satisfaction and acoustical parameters, and the results are summarized in Table 5. Among the four single number quantities presented in ISO 3382–3, satisfaction exhibits the largest negative correlation with $L_{p,A,S,4m}$ (r = –0.62, p < 0.01) and a distinctly positive correlation with $D_{2,S}$ (r = 0.41, p < 0.01). However, no significant correlation with r_D (r = –0.01, p = 0.90) is observed. Furthermore, satisfaction exhibits a strong negative correlation with RT (r = –0.53, p < 0.01), indicating the degree of reverberation of the office sound environment, and a strong positive correlation with D_{50} (r = 0.51, p < 0.01). The other preference attributes, excluding annoyance, also exhibit similar tendencies to satisfaction. In contrast, annoyance shows correlations with the acoustic parameters that are opposite to those of the other preference indicators because it evaluates the negative aspect of the OPO sound environment, unlike the other indicators. Therefore, among the acoustical parameters related to speech privacy, the preference for the OPO sound environment generally shows negative correlations with $L_{p,A,S,4m}$, RT, and the STI but a positive correlation with D_{50}.

For stakeholders such as general architects and designers who are not researchers in the academic field of indoor building acoustics, considering various acoustical parameters simultaneously when designing an OPO sound environment is challenging. Therefore, it is crucial to determine the physical acoustic performance that must be ensured first. Consequently, linear regression analysis was conducted to investigate the contributions of various acoustical parameters to the OPO sound environment and then to select indicators to be utilized first in the design stage. Six physical acoustic parameters were specified as independent parameters, whereas the satisfaction derived by PCA was specified as the dependent parameter. A collinearity test was performed, and the parameters with variance inflation factors of 10 or higher and tolerances of 0.1 or lower were excluded because they were considered to violate collinearity. The results are summarized in Table 6. The total R^2 of the regression model indicates statistical significance ($F(3,440) = 95.923$, p < 0.01; R^2 = 0.395). The R^2 value obtained was 0.395; thus, and 39.5% of the satisfaction variance was predicted or explained by $L_{p,A,S,4m}$, $D_{2,S}$, and r_D. Although the standardized coefficients of $L_{p,A,S,4m}$ and $D_{2,S}$ (–0.27 and –0.12, respectively) are statistically significant, r_D is not. In particular,

Table 5
Pearson correlations between acoustical parameters and preference attributes.

Attributes	Acoustical parameters					
	$L_{p,A,S,4m}$	$D_{2,S}$	r_D	STI	RT	D_{50}
Satisfaction	−0.62**	0.41**	−0.01	−0.18**	−0.53**	0.51**
Willingness	−0.67**	0.45**	0.01	−0.19**	−0.58**	0.57**
Annoyance	0.56**	−0.37**	−0.01	0.15**	0.47**	−0.44**
Privacy	−0.47**	0.31**	−0.05	−0.16**	−0.38**	0.37**
Immersion	−0.57**	0.37**	−0.01	−0.16**	−0.48**	0.46**
Pleasantness	−0.52**	0.34**	0.03	−0.12**	−0.46**	0.44**
Comfort	−0.53**	0.37**	−0.03	−0.17**	−0.46**	0.44**

Table 6
Summary of the linear regression of satisfaction with acoustical parameters (R^2 = 0.395, p < 0.01).

Acoustical parameters	Unstandardized coefficients	t	p	Correlations			ΔR^2
				Zero-order	Partial	Part	
(Constant)	14.04	9.64	<0.01				
$L_{p,A,S,4m}$	−0.27	−11.71	<0.01	−0.62	−0.49	−0.43	0.19
$D_{2,S}$	−0.12	−2.39	<0.05	0.41	−0.11	−0.09	0.01
r_D	−0.03	−1.38	0.17	−0.01	−0.07	−0.05	0.01

when the independent contributions of each independent parameter in predicting or explaining the satisfaction variant are compared, $L_{p,A,S,4m}$ exhibits the highest contribution of 0.19, whereas $D_{2,S}$ shows a very low contribution of 0.01. Therefore, $L_{p,A,S,4m}$ must be considered first in the design stage owing to its highest influence on the determination of satisfaction of the OPO sound environment among the acoustical parameters. This result is somewhat different from the findings of Lee and Jeon [38] and Virjonen et al. [40], who found that r_D showed the highest correlation with speech privacy. However, the previous studies examined correlations considering intelligibility or privacy of speech rather than preferences for the OPO sound environment, which could explain why the influence of r_D was smaller than those of the other indicators.

3.2. Effect of noise sensitivity on satisfaction

According to the results discussed above, the changes in participant satisfaction were examined based on $L_{p,A,S,4m}$. The percentage of participants with a factor score higher than zero, the median factor score, was calculated, and the results are summarized in Fig. 4. Probit regression analysis is a regression model that analyzes binomial response variables. It is used to analyze subjective responses to indoor noise owing to its advantage of straightening the S-shaped curve using normal distribution probability [60–63]. Therefore, probit regression analysis was conducted to fit the satisfaction changes, and the results are illustrated in Fig. 4. The probit model revealed statistical significance (p < 0.01). Consequently, it was found that to satisfy 50% of the participants group, decreasing $L_{p,A,S,4m}$ to below 48.2 dB is desirable. Further, when $L_{p,A,S,4m}$ was divided into sections of 25% by the percentage of satisfaction, it was necessary to decrease $L_{p,A,S,4m}$ to 46.3 dB or less to satisfy 75% (class A) of all participants, and to decrease $L_{p,A,S,4m}$ to 50.1 dB or less to satisfy 25% (class C).

To compare the satisfaction changes according to sensitivity to noise, the total participant group was classified by sensitivity as follows: seven participants in the top 20% (high), 23 participants in the middle 60% (normal), and seven participants in the bottom 20% (low). The results obtained after the probit regression analysis for satisfaction changes by sensitivity group according to the change in $L_{p,A,S,4m}$ are shown in Fig. 5. Each probit model exhibits statistical significance (p < 0.01). In general, the group that was more sensitive to noise had a greater need for an environment with

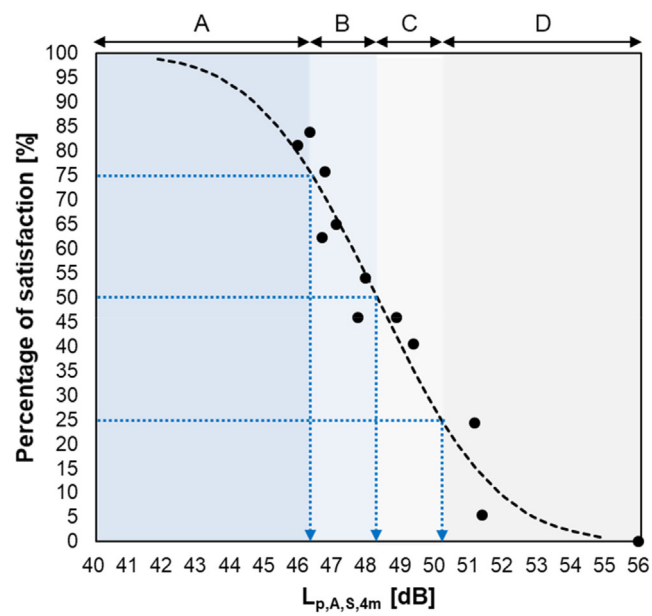

Fig. 4. Percentage of satisfaction according to $L_{p,A,S,4m}$.

a lower $L_{p,A,S,4m}$. The $L_{p,A,S,4m}$ values that satisfy 50% of all participants in this study are 46.2, 48.4, and 49.4 for the high, normal, and low sensitivity groups, respectively. It can be observed that the difference between the normal and low sensitivity groups is small, at approximately 1.0 dB, but the differences between the high sensitivity group and the normal and low sensitivity groups are somewhat large, at 2.2 and 3.2 dB, respectively.

4. Discussion

Based on the correlations between $L_{p,A,S,4m}$ and satisfaction examined above, the four classes corresponding to the 25% sections of satisfaction are summarized in Fig. 6. The findings of Virjonen et al. [40], which form the basis of the target value proposed in ISO 3392–3 Annex A, are also shown in this figure. Considering the classification criteria proposed in this study, it is necessary to achieve a sound environment condition of 46 dB or less, which is

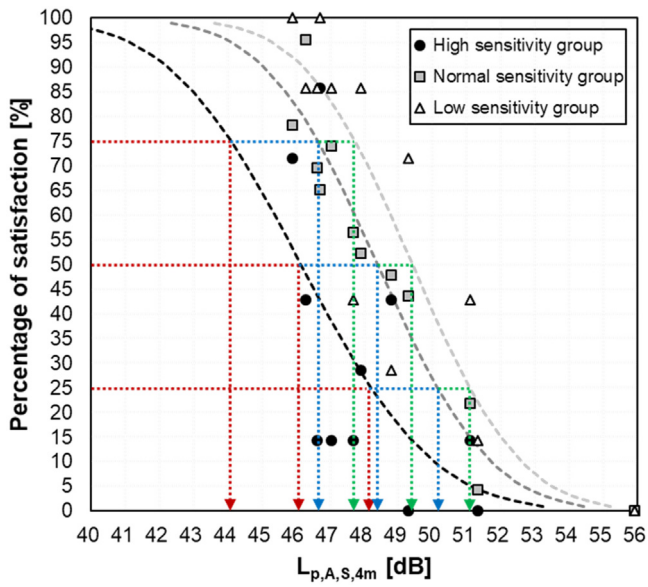

Fig. 5. Percentage of satisfaction according to sensitivity group.

slightly stricter than the criterion of 48 dB suggested in the ISO standard to ensure the psychological satisfaction of workers in an office environment. The reason for this difference is that previous studies presented the criteria based on case study data for the physical sound environments of various OPOs; thus, limitations in reflecting the emotions of workers after actually experiencing the sound environment are expected. Furthermore, this difference appears in the size of the class intervals. When the class interval was set based on the participant responses in this study, it was necessary to select a more sensitive class interval of 2 dB than the existing classification criterion of 3 dB. However, this study only examined $D_{2,S}$ and r_D in the ranges of 2.5–8.1 and 6.1–13.2 m, respectively. Thus, additional research on offices under various conditions is required to derive more generalized conclusions.

To compare sound environment class intervals according to noise sensitivity, the class A criterion of 48 dB, which can satisfy

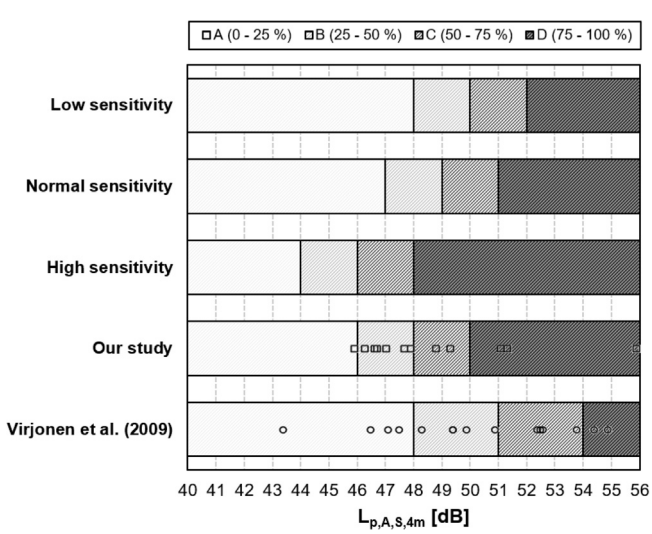

Fig. 6. Classification criteria based on noise sensitivity (the rectangular symbols indicate the sound stimuli used in this study, and the circles represent the measurement results of a previous study [40]).

Table 7
Summary of ANOVA analysis results with satisfaction differences according to $L_{p,A,S,4m}$. (* means $p < 0.05$, and ** means $p < 0.01$).

Site	Δ $L_{p,A,S,4m}$ [dB]	Δ $D_{2,S}$ [dB]	Δ r_D [m]	Case comparison	Δ Factor
Office I	0.3	−0.2	0.0	4 ↔ 1	−0.18
	0.4	−0.7	0.7	2 ↔ 4	−0.15
	0.6	−0.4	0.3	5 ↔ 2	−0.39
	0.7	−0.9	0.7	2 ↔ 1	−0.33
	1.1	−1.1	1.0	5 ↔ 4	−0.54
	1.4	−1.3	1.0	5 ↔ 1	−0.72**
	2.0	−0.8	0.0	6 ↔ 3	−0.71**
	2.7	−2.1	1.7	3 ↔ 4	−0.70*
	3.0	−2.3	1.7	3 ↔ 1	−0.88**
	3.6	−1.8	0.8	6 ↔ 5	−0.87**
	4.3	−2.2	1.0	6 ↔ 2	−1.26**
	4.7	−2.9	1.7	6 ↔ 4	−1.41**
	5.0	−3.1	1.7	6 ↔ 1	−1.59**
Office II	0.8	−0.2	0.2	10 ↔ 7	−0.24
	0.9	−0.2	1.2	11 ↔ 8	−0.19
	1.3	−0.3	1.0	8 ↔ 10	−0.40
	2.1	−0.5	1.2	8 ↔ 7	−0.64*
	2.9	−0.7	2.4	11 ↔ 7	−0.83**
	3.1	−0.8	3.1	9 ↔ 8	−0.66*
	4.4	−1.1	4.1	9 ↔ 10	−1.05**
	4.8	−1.4	−7.1	12 ↔ 9	−0.94**
	5.2	−1.3	4.3	9 ↔ 7	−1.30**
	7.1	−2.0	−5.2	12 ↔ 11	−1.41**
	8.0	−2.2	−4.0	12 ↔ 8	−1.60**
	9.2	−2.5	−3.0	12 ↔ 10	−2.00**
	10.0	−2.7	−2.8	12 ↔ 7	−2.24**

75% or more of the low sensitivity group, was considered the minimum design condition to satisfy 25% of participants, which corresponds to class C, in the high sensitivity group. Thus, a large difference exists. Moreover, when compared with the existing criteria of ISO 3382–3, the target value of 48 dB was the same as the $L_{p,A,S,4m}$ criterion derived from the low sensitivity group, further confirming the need for stricter design criteria than the existing criteria to satisfy the normal and high sensitivity groups, corresponding to 80% of all participants. However, in this study, only the change in $L_{p,A,S,4m}$ was considered; as shown in ISO 3382–3, speech privacy in the OPO is related to various acoustic parameters such as r_D and $D_{2,S}$, and it is necessary to consider them together.

To clarify the basis for selecting 2 dB as the class interval, the statistical significance of the differences in satisfaction response according to the change in the $L_{p,A,S,4m}$ condition was examined through analysis of variance (ANOVA), and the results are shown in Table 7. To verify whether the noticeable differences for both offices were the same, each office was analyzed separately. However, because the satisfaction data distribution according to each condition of $L_{p,A,S,4m}$ did not satisfy the homoscedasticity condition, it was analyzed using the Welch test. Fourteen differences of $L_{p,A,S,4m}$ in the range of 0.3–5.0 are observed in Office I according to six combinations of $L_{p,A,S,4m}$ used in this experiment, and the satisfaction response changes significantly from a difference of 1.4 dB or more (df1 = 5, df2 = 100.073, $p < 0.01$). For Office II, $L_{p,A,S,4m}$ changes from 0.8 to 10.0 dB depending on the stimulus combination and the satisfaction response changes significantly at 2.1 dB or higher (df1 = 5, df2 = 92.033, $p < 0.01$). Furthermore, in Office II, the satisfaction responses does not change significantly for an $L_{p,A,S,4m}$ difference of 1.3 dB. Considering that satisfaction shows a significant change only for an $L_{p,A,S,4m}$ difference of 1.4 dB or higher in Office I, it is evident that the difference that can change the satisfaction of participants is approximately 1.4 dB. Therefore, it can be inferred that a class interval of 2 dB is appropriate. Moreover, when the existing noise criteria for indoor environments

outlined by Rasmussen [64] are examined, it is evident that the class intervals of 3–5 dB are selected in the design criteria for noises inside buildings (e.g., impact sounds and airborne sounds). Thus, when OPOs are designed in terms of speech privacy, as in this study, a design criterion of 2 dB or lower is appropriate considering the functional characteristics of OPOs in which workers concentrate more on and react more sensitively to speech than in other indoor spaces. However, as shown in Table 7, because $D_{2,S}$ and r_D also change when $L_{p,A,S,4m}$ changes, 1.4 dB is somewhat limited as the minimum difference that causes the subjective response difference.

5. Conclusions

In this study, the preferences of participants for changes in the physical sound environment of OPOs using various attributes were evaluated, quantified by the single measure of "satisfaction," and a classification criterion for OPO sound environments was proposed by investigating the correlations of satisfaction with acoustical parameters. The findings of this study are as follows.

1. As a result of examining the relationship between speech privacy-related parameters and preferences in the aspect of satisfaction with OPO, we found that $L_{p,A,S,4m}$, among the physical acoustic parameters analyzed, was the most reliable variable in determining satisfaction;
2. When selecting the classification criteria of $L_{p,A,S,4m}$ in the satisfaction aspect, we found that it was desirable to select them in 2 dB intervals, considering the acoustic environment of OPO;
3. To ensure the satisfaction of workers, a sound environment condition of 46 dB or less must be secured, which is slightly stricter than the criteria of 48 dB from the existing ISO 3382–3;
4. Satisfaction with OPO considerably varies depending on the noise sensitivity. In particular, strict acoustic design criteria should be applied to satisfy normal and high sensitivity.

The findings and classification criteria in this study are expected to be used as base data for designing pleasant OPOs in the future. Despite these findings, this study has the following limitations. First, as only the test participants in their 20 s were evaluated, and college students were targeted in this study, there was a limitation in generalizing the results for all age and occupational groups. Second, sufficient variations of evaluation stimuli were not ensured for the $D_{2,S}$ and r_D indicators representing the characteristics of the OPO sound environment in addition to $L_{p,A,S,4m}$. As shown in previous studies, $L_{p,A,S,4m}$ can be affected by the surrounding furniture or walls. Thus, follow-up studies must complement the design criteria by considering $D_{2,S}$ and r_D, which indicate the overall characteristics of the space [40]. Third, this study only used auditory stimuli and did not include other stimuli (e.g., visual and tactile stimuli) that participants actually experience in their office spaces. Despite such limitations, this study is meaningful as it suggests design criteria for the sound environment of an OPO from the perspective of human perception (specifically in terms of preference) rather than the traditional method of presenting grades based on physical acoustic parameters. In addition, although task performance was not considered in this study, it is more appropriate to evaluate the sound environment of an OPO while the workers are performing some practical task. Therefore, it is necessary to examine the correlation with work performance in future studies. Nevertheless, this study is meaningful in that it considered the existing sound environment grading standards based on the effect of the SPL difference of speech on satisfaction in laboratory conditions, under which factors other than sound were controlled.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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